



2022 YEAR 12 Assessment Task 1

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 1 hour
- Write using black pen.
- Calculators approved by NESA may be used.
- In Questions 6 – 8, show relevant mathematical reasoning and/or calculations.
- Answer all questions in the booklet provided.

Total marks: 34

Section I – 5 marks (pages 2 – 3)

- Attempt Questions 1 – 5
- Allow about 8 minutes for this section.

Section II – 29 marks (pages 4 – 6)

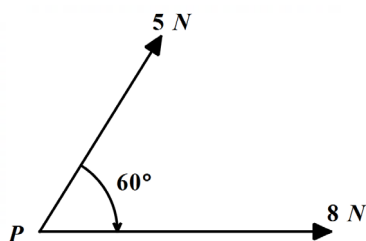
- Attempt Questions 6 – 8
- Allow about 52 minutes for this section.

Section I - 5 marks**Answer on the multiple choice answer sheet**

1. The points P , Q and R lie on a straight line. The vector $\overrightarrow{PQ} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$ and the vector $\overrightarrow{PR} = 2\overrightarrow{PQ}$. The coordinates of P are $(2, -2)$. What are the coordinates of R ?
- (A) $(-2, 4)$
- (B) $(-6, 8)$
- (C) $(-6, 10)$
- (D) $(6, -10)$

2. Consider the vectors $\underline{a} = 2\underline{i} + 3\underline{j}$, $\underline{b} = -3\underline{i} + 2\underline{j}$ and $\underline{c} = 2\underline{i} - \underline{j}$. Which of the following vectors is parallel to $\underline{a} + \underline{b} + \underline{c}$?
- (A) $-2\underline{i} - 6\underline{j}$
- (B) $2\underline{i} + 8\underline{j}$
- (C) $2\underline{i} - 6\underline{j}$
- (D) $2\underline{i} + 6\underline{j}$

3. Forces of magnitude 8 N and 5 N act on a particle P . The angle between the directions of the two forces is 60° as shown in the diagram.

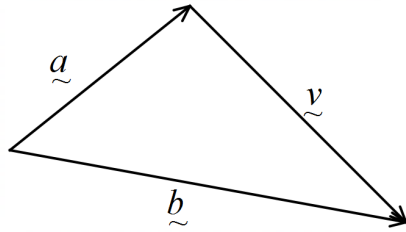


Which of the following is correct magnitude and direction of the resultant force acting on P ?

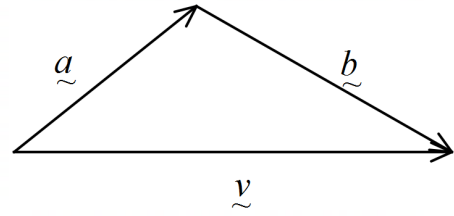
- (A) 11.36 N , $22^\circ 25'$ to the horizontal
- (B) 11.36 N , $67^\circ 35'$ to the horizontal
- (C) 12.58 N , $22^\circ 25'$ to the horizontal
- (D) 12.58 N , $67^\circ 35'$ to the horizontal

4. Which diagram below shows the vector $\underline{v} = \underline{b} - \underline{a}$?

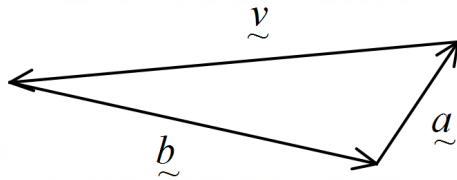
A



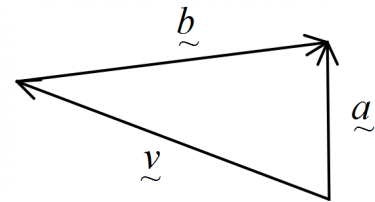
B



C



D



5. Consider the two non-zero vectors, $\underline{u}$ and $\underline{v}$, with the following properties:

- $\underline{v} = k\underline{u}$, where k is a constant and
- $\underline{u} \cdot \underline{v} < 0$.

Which of the following statements is true?

- (A) $\underline{u} \perp \underline{v}$
- (B) $\underline{u} \parallel \underline{v}$ in the same direction
- (C) $\underline{u} \parallel \underline{v}$ in opposite directions
- (D) The angle between the vectors $\underline{u}$ and $\underline{v}$ is acute

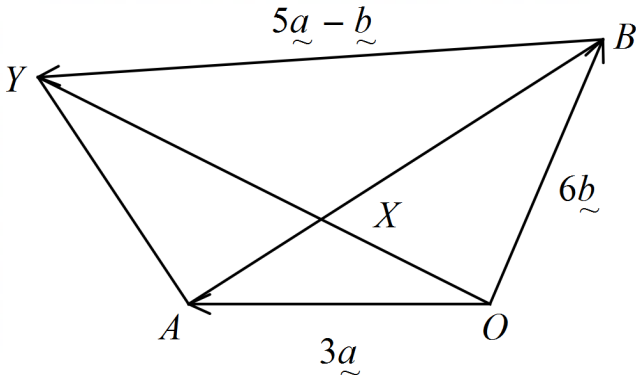
End of Section I

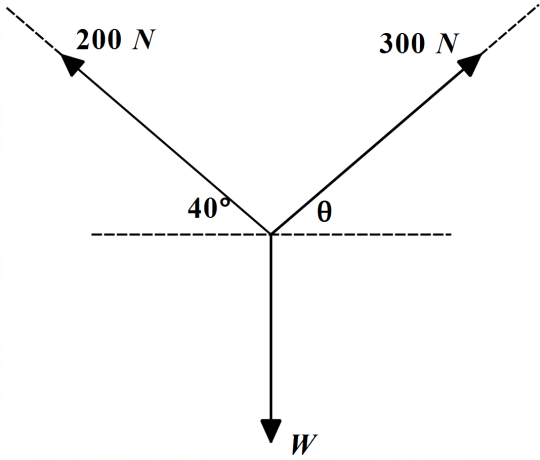
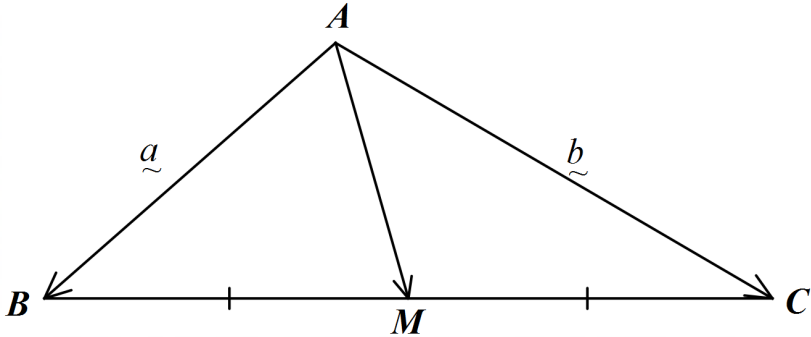
Section II – Extended Response

Attempt questions 6-8.

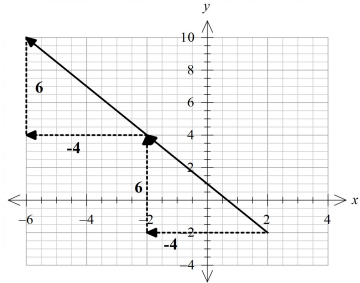
Answer each question on the appropriate pages of the answer booklet, showing all necessary working.

Question 6 (11 marks)	Marks
(a) Consider the vectors $\overrightarrow{OP} = -2\hat{i} + \hat{j}$ and $\overrightarrow{OQ} = 4\hat{i} - 3\hat{j}$. (i) Find the vector $\underline{u} = \overrightarrow{PQ}$. (ii) Given that $\underline{v} = -\hat{i} + 2\hat{j}$, find $\underline{u} \cdot \underline{v}$. (iii) Find $\text{proj}_{\underline{u}} \underline{v}$.	1 1 2
(b) Consider $\triangle ABC$ below where $\overrightarrow{AB} = \underline{c}$, $\overrightarrow{BC} = \underline{a}$, $\overrightarrow{AC} = \underline{b}$ and $\angle ABC = 90^\circ$. Let M be the midpoint of AC. (i) Explain why $\overrightarrow{MB} = \underline{c} - \frac{\underline{b}}{2}$. (ii) Find the expression for $\overrightarrow{MC}$ in terms of $\underline{a}$, $\underline{b}$ and $\underline{c}$.	1 1
(c) (i) Prove by mathematical induction that for all integers $n \geq 1$, $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{1}{6}n(n+1)(2n+1).$ (ii) Use the above result to show that $2^2 + 4^2 + 6^2 + \dots + 100^2 = 171700$. (iii) Hence evaluate $1^2 + 3^2 + 5^2 + \dots + 99^2$.	3 1 1
End of Question 6	

Question 7 (8 marks)	Marks
a) Two vectors are given by $\underline{a} = 3\underline{i} - m\underline{j}$ and $\underline{b} = -2\underline{i} + n\underline{j}$, where m and n are real with $m > 0$. If $\underline{a} = 5$ and $\underline{a}$ is perpendicular to $\underline{b}$, find the values of m and n.	2
b) In the quadrilateral $OAYB$, $\overrightarrow{OA} = 3\underline{a}$, $\overrightarrow{OB} = 6\underline{b}$ and $\overrightarrow{BY} = 5\underline{a} - \underline{b}$. AB intersects OY at X.  (i) Express $\overrightarrow{AB}$ in terms of $\underline{a}$ and $\underline{b}$. (ii) X is the point on AB such that $\overrightarrow{AX} = \frac{1}{2}\overrightarrow{XB}$. Prove that $\overrightarrow{OX} = \frac{2}{5}\overrightarrow{OY}$.	1 2
c) Use the method of mathematical induction to prove that $2^{3n} - 3^n$ is divisible by 5 for all integer, $n \geq 1$.	3
End of Question 7	

Question 8 (10 marks)	Marks
a) The diagram shows the weight force, W, that pulls straight down on an object. Two forces of 200 N and 300 N are exerted by cables as shown. If the resultant force is zero, find the size of θ to the nearest degree and magnitude of W to 2 decimal places. 	3
b) Use mathematical induction to prove that for integers $n \geq 1$, $1 + 2 \times \frac{1}{2} + 3 \times \left(\frac{1}{2}\right)^2 + 4 \times \left(\frac{1}{2}\right)^3 + \dots + n \times \left(\frac{1}{2}\right)^{n-1} = 4 - \left(\frac{n+2}{2^{n-1}}\right).$	3
c) In the diagram, M is the midpoint of the side BC of triangle ABC. Let $\overrightarrow{AB} = \underline{a}$ and $\overrightarrow{AC} = \underline{b}$.  (i) Express the vector $\overrightarrow{AM}$ and $\overrightarrow{CM}$ in terms of $\underline{a}$ and $\underline{b}$. (ii) Show that $\overrightarrow{AM} ^2 + \overrightarrow{CM} ^2 = \frac{1}{2}(\overrightarrow{AB} ^2 + \overrightarrow{AC} ^2)$.	2 2
End of Exam	

HSC Year12 Extension 1 Task 1 2022(suggested marking guidelines)

	Solutions	Mks	Comments
1	This problem could be best solved graphically  We are given that the coordinates of P are (2, -2). The vector $\overrightarrow{PQ} = (-4, 6)$, so Q has coordinates (-2, 4). Given $\overrightarrow{PR} = 2\overrightarrow{PQ}$, $\overrightarrow{QR} = \overrightarrow{PQ}$ so R will have coordinates (-6, 10).	1	C
2	$\underline{a} + \underline{b} + \underline{c} = 2\underline{i} + 3\underline{j} - 3\underline{i} + 2\underline{j} + 2\underline{i} - \underline{j}$ $= \underline{i} + 4\underline{j}$ $2\underline{i} + 8\underline{j} = 2(\underline{i} + 4\underline{j})$ $= 2(\underline{a} + \underline{b} + \underline{c})$ then gives option B	1	B
3	Horizontal components: $x = 5 \cos 60^\circ + 8$ Vertical components: $y = 5 \sin 60^\circ + 0$ Magnitude = $\sqrt{x^2 + y^2}$ $= \sqrt{(5 \cos 60^\circ + 8)^2 + (5 \sin 60^\circ)^2}$ $= 11.3578..$ $= 11.36 \text{ N}$ Direction : $\tan \theta = \frac{y}{x}$ $= \frac{5 \sin 60^\circ}{5 \cos 60^\circ + 8}$ $\therefore \theta = 22^\circ 24' 39.28''$ $= 22^\circ 25'$ (This step is not necessary as θ must be less than 60°, but you can use it as a check)	1	A
4	A. $\underline{a} + \underline{v} = \underline{b}$ $\underline{v} = \underline{b} - \underline{a}$ B. $\underline{a} + \underline{b} = \underline{v}$ C. $\underline{b} + \underline{a} = -\underline{v}$ $\underline{v} = -\underline{a} - \underline{b}$ D. $\underline{v} + \underline{b} = \underline{a}$ $\underline{v} = \underline{a} - \underline{b}$	1	A

	Solutions	Mks	Comments
5	$\underline{u} + \underline{v} = k \underline{u}$ means $\underline{u}$ and $\underline{v}$ are parallel vectors $\underline{u} \cdot \underline{v} < 0$ means angle between them is $> 90^\circ \therefore$ in opposite direction	1	C
6a(i)	$\overrightarrow{OP} = -2\hat{i} + \hat{j} \quad \overrightarrow{OQ} = 4\hat{i} - 3\hat{j}$ $\underline{u} = \overrightarrow{PO} + \overrightarrow{OQ}$ $= -(\overrightarrow{OP}) + \overrightarrow{OQ}$ $= -(-2\hat{i} + \hat{j}) + (4\hat{i} - 3\hat{j})$ $= 6\hat{i} - 4\hat{j}$	1	1 mark • correct answer.
6a(ii)	$\underline{u} \cdot \underline{v} = (6\hat{i} - 4\hat{j}) \cdot (-\hat{i} + 2\hat{j})$ $= 6 \times -1 + (-4) \times 2$ $= -14$	1	1 mark • correct solution.
6a(iii)	$\text{proj}_{\underline{u}} \underline{v} = \frac{\underline{u} \cdot \underline{v}}{ \underline{u} } \underline{u}$ $= \frac{-14}{(\sqrt{6^2 + (-4)^2})^2} (6\hat{i} - 4\hat{j})$ $= \frac{-84}{52} \hat{i} + \frac{56}{52} \hat{j}$ $= \frac{-21}{13} \hat{i} + \frac{14}{13} \hat{j}$	2	2 mark • correct solution. 1 mark Adequately using the formula for projection.
6b(i)	We know that $\overrightarrow{AB} = \underline{c}$. Since M is the midpoint of $\overrightarrow{AC}$, then $\overrightarrow{AM} = \frac{\underline{b}}{2}$ $\overrightarrow{MB} = \overrightarrow{MA} + \overrightarrow{AB}$ $= -\frac{\underline{b}}{2} + \underline{c} \text{ (as required)}$	1	1 mark • correct explanation with vector notation.
6(b)(ii)	$\overrightarrow{MC} = \overrightarrow{MB} + \overrightarrow{BC}$ $= \underline{c} - \frac{\underline{b}}{2} + \underline{a}$	1	1 mark • correct answer.
6c(i)	Let $n = 1$ $LHS = 1^2$ $RHS = \frac{1}{6} \times 1 \times (1+1) \times (2 \times 1 + 1)$ $= \frac{1}{6} \times 6$ $= 1 = LHS$ $\therefore$ true for $n = 1$ Assume true for $n = k$	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof

	Solutions	Mks	Comments
	i.e. $1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{1}{6}k(k+1)(2k+1)$ Prove true for $n = k + 1$ $1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{1}{6}(k+1)(k+2)(2k+3)$ <i>LHS</i> $1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2$ $= \frac{1}{6}(k)(k+1)(2k+1) + (k+1)^2 \text{ (from assumption)}$ i.e. $= \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)]$ $= \frac{1}{6}(k+1)(2k^2 + 7k + 6)$ $= \frac{1}{6}(k+1)(k+2)(2k+3)$ $= RHS$ $\therefore$ if true for $n = k$, then true for $n = k + 1$. But it is true for $n = 1$ Hence it is true for $n = 1 + 1 = 2$, since true for $n = 2$, it is true for $n = 3$ and so on, then it is true for all positive integer n.		4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 marks • Successfully does 2 of the 4 key parts
6c(ii)	$2^2 + 4^2 + 6^2 + \dots + 100^2 = 2^2(1^2 + 2^2 + 3^2 + \dots + 50^2)$ $= 4 \times \frac{1}{6} \times 50 \times (50+1)(2 \times 50 + 1) \text{ using the result from part(i)}$ $= 171700$	1	1 mark • Correct solution.
6c(iii)	$1^2 + 3^2 + 5^2 + \dots + 99^2$ $= (1^2 + 2^2 + 3^2 + \dots + 100^2) - (2^2 + 4^2 + 6^2 + \dots + 100^2)$ $= \frac{1}{6} \times 100 \times (100+1)(2 \times 100 + 1) - 171700$ $= 166650$	1	1 mark • Correct solution.
7a	$\underline{a} = 3\underline{i} - m\underline{j}$ $\underline{b} = -2\underline{i} + n\underline{j}$ $ \underline{a} = 5 = \sqrt{3^2 + m^2} \quad \underline{a} \perp \underline{b} \Rightarrow \underline{a} \cdot \underline{b} = 0$ $m^2 = 16 \quad \therefore 3(-2) - 4n = 0$ $\text{as } m > 0 \therefore m = 4 \quad 4n = -6$ $n = -\frac{3}{2}$	2	2 mark • correct solution. 1 mark • either finds n or m.
7b(i)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}$ $= -\overrightarrow{OA} + \overrightarrow{OB}$ $= 6\underline{b} - 3\underline{a}$	1	1 mark • Correct answer.

	Solutions	Mks	Comments
7b (ii)	$\overrightarrow{OX} = 3\vec{a} + \frac{1}{3}(6\vec{b} - 3\vec{a})$ $= 2\vec{a} + 2\vec{b}$ $\frac{2}{5}\overrightarrow{OY} = \frac{2}{5}(6\vec{b} + 5\vec{a} - \vec{b})$ $= 2\vec{a} + 2\vec{b}$ $\therefore \overrightarrow{OX} = \frac{2}{5}\overrightarrow{OY}.$	2	2 marks • correct solution. 1 mark • significant progress towards solution.
7c	<u>Test for $n = 1$</u> $2^{3(1)} - 3 = 8 - 3 = 5$ $\therefore$ divisible by 5 $\therefore$ true for $n = 1$. <u>Assume true for $n = k$</u> i.e $2^{3k} - 3^k = 5P$, where P is a positive integer Prove true for $n = k + 1$ i.e $2^{3(k+1)} - 3^{k+1} = 5Q$ where Q is a positive integer $2^{3k} \cdot 2^3 - 3^k \cdot 3$ $(5P + 3^k) \cdot 8 - 3 \cdot 3^k$ LHS= $40P + 8 \cdot 3^k - 3 \cdot 3^k$ (from assumption) $40P - 5 \cdot 3^k$ $5(8P - 3^k)$ $5Q \text{ where } Q \text{ is a positive integer}$ Which is divisible by 5. Hence the statement is true for $n = k + 1$ if it is true for $n = k$. By Induction, since it is true for $n = 1$, then it is also true for $n = 2, 3, 4, \dots$ all positive integer of n	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 marks • Successfully does 2 of the 4 key parts Note: Without using the assumption, it is not the process of mathematical induction

	Solutions	Mks	Comments
8a	Equating up and down and left and right $200 \sin 40 + 300 \sin \theta = w \quad (1)$ $300 \cos \theta - 200 \cos 40 = 0 \quad (2)$ $300 \cos \theta = 200 \cos 40$ $\theta = \cos^{-1} \left(\frac{2}{3} \cos 40 \right)$ $\theta = 59.28 \dots$ sub $\theta = 59.28 \dots$ into (1) $200 \sin 40 + 300 \sin 59.28 \dots = w$ $w = -386.49 \dots$ $\therefore \theta = 59.28$ and $w = 386.49$ (to 2 dp)	3	3 mark • correct solution. 2 mark • Correct θ. 1 mark • one correct horizontal/vertical equation.
8b	Let $n = 1$ $LHS = 1$ $RHS = 4 - \frac{1+2}{2^{1-1}}$ $= 4 - 3$ $= 1 = LHS$ $\therefore$ true for $n = 1$ Assume true for $n = k$ i.e. $1 + 2 \times \frac{1}{2} + 3 \times \left(\frac{1}{2}\right)^2 + \dots + k \left(\frac{1}{2}\right)^{k-1} = 4 - \frac{k+2}{2^{k-1}}$ Prove true for $n = k + 1$ $1 + 2 \times \frac{1}{2} + 3 \times \left(\frac{1}{2}\right)^2 + \dots + k \left(\frac{1}{2}\right)^{k-1} + (k+1) \left(\frac{1}{2}\right)^{k+1-1} = 4 - \frac{(k+1)+2}{2^{k+1-1}} = 4 - \frac{k+3}{2^k}$ LHS $1 + 2 \times \frac{1}{2} + 3 \times \left(\frac{1}{2}\right)^2 + \dots + k \left(\frac{1}{2}\right)^{k-1} + (k+1) \left(\frac{1}{2}\right)^k$ $= 4 - \frac{(k+2)}{2^{k-1}} + (k+1) \left(\frac{1}{2}\right)^k \text{ (from assumption)}$ $= 4 - \frac{(k+2)}{2^{k-1}} + (k+1) \frac{1}{2^k}$ $= 4 + \frac{-2(k+2) + k + 1}{2^k}$ $= 4 + \frac{-k - 3}{2^k}$ $= 4 - \frac{k+3}{2^k} = RHS$ If it is true for $n = k$, then it is also true for $n = k + 1$. By Induction, since it is true for $n = 1$, then it is also true for $n = 2, 3, 4 \dots$ all positive integer of n	3	There are 4 key parts of the induction; 1. Proving the result true for $n = 1$ 2. Clearly stating the assumption and what is to be proven 3. Using the assumption in the proof 4. Correctly proving the required statement 3 marks • Successfully does all of the 4 key parts 2 marks • Successfully does 3 of the 4 key parts 1 marks • Successfully does 2 of the 4 key parts Note: Without using the assumption, it is not the process of mathematical induction

	Solutions	Mks	Comments
8c(i)	$\overrightarrow{AM} = \underline{a} = \overrightarrow{BM} \quad (1)$ <i>also</i> $\overrightarrow{AM} = \underline{b} + \overrightarrow{CM}$ and as $\overrightarrow{CM} = -\overrightarrow{BM}$ then $\overrightarrow{AM} = \underline{b} - \overrightarrow{BM} \quad (2)$ <i>By adding (1) and (2)</i> $2\overrightarrow{AM} = \underline{a} + \underline{b}$ $\overrightarrow{AM} = \frac{1}{2}(\underline{a} + \underline{b})$ $\overrightarrow{CM} = -\underline{b} + \overrightarrow{AM}$ $\overrightarrow{CM} = -\underline{b} + \frac{1}{2}(\underline{a} + \underline{b})$ $= -\underline{b} + \frac{1}{2}\underline{a} + \frac{1}{2}\underline{b}$ $= \frac{1}{2}\underline{a} - \frac{1}{2}\underline{b}$ $= \frac{1}{2}(\underline{a} - \underline{b})$	2	2 mark - correct solution. 1 mark - finds the correct expression for $\overrightarrow{AM}$ or $\overrightarrow{CM}$..
8c(ii)	$= \overrightarrow{AM} ^2 + \overrightarrow{CM} ^2$ $= \frac{1}{2}(\underline{a} + \underline{b}) \cdot \frac{1}{2}(\underline{a} + \underline{b}) + \frac{1}{2}(\underline{a} - \underline{b}) \cdot \frac{1}{2}(\underline{a} - \underline{b})$ $= \frac{1}{4}(\underline{a} + \underline{b}) \cdot (\underline{a} + \underline{b}) + \frac{1}{4}(\underline{a} - \underline{b}) \cdot (\underline{a} - \underline{b})$ LHS $= \frac{1}{4}(\underline{a} ^2 + 2\underline{a} \cdot \underline{b} + \underline{b} ^2 + \underline{a} ^2 - 2\underline{a} \cdot \underline{b} + \underline{b} ^2)$ $= \frac{1}{4}(2 \underline{a} ^2 + 2 \underline{b} ^2)$ $= \frac{1}{2}(\underline{a} ^2 + \underline{b} ^2)$ $= \frac{1}{2}(\overrightarrow{AB} ^2 + \overrightarrow{AC} ^2)$ $= RHS$	2	2 mark - correct solution. 1 mark - finds the correct expression for $\overrightarrow{AM} ^2 + \overrightarrow{CM} ^2$ in terms of $\underline{a}$ and $\underline{b}$.